

Mechanical Properties of Fluids –II

Topics Covered:

- Atmospheric Pressure
- Absolute Pressure & Gauge Pressure
- Hydrostatic Paradox

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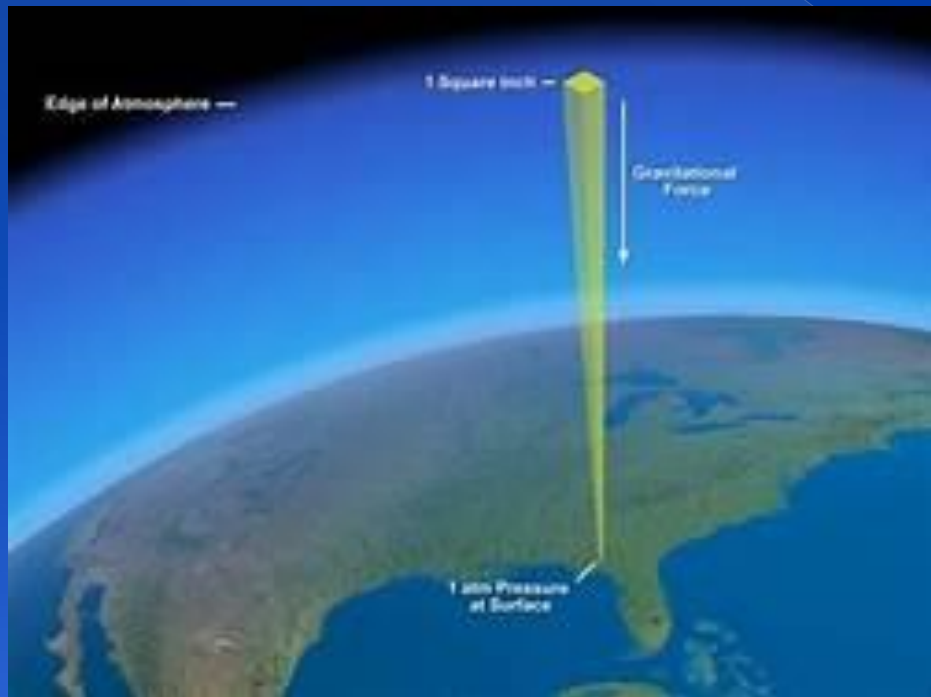


2.3.2 Atmospheric Pressure:

- Earth's atmosphere is made up of a fluid, namely, air. It exerts a downward force due to its weight.

The pressure due to this force is called atmospheric pressure.

Thus, at any point, the atmospheric pressure is the weight of a column of air of unit cross section starting from that point and extending to the top of the atmosphere.

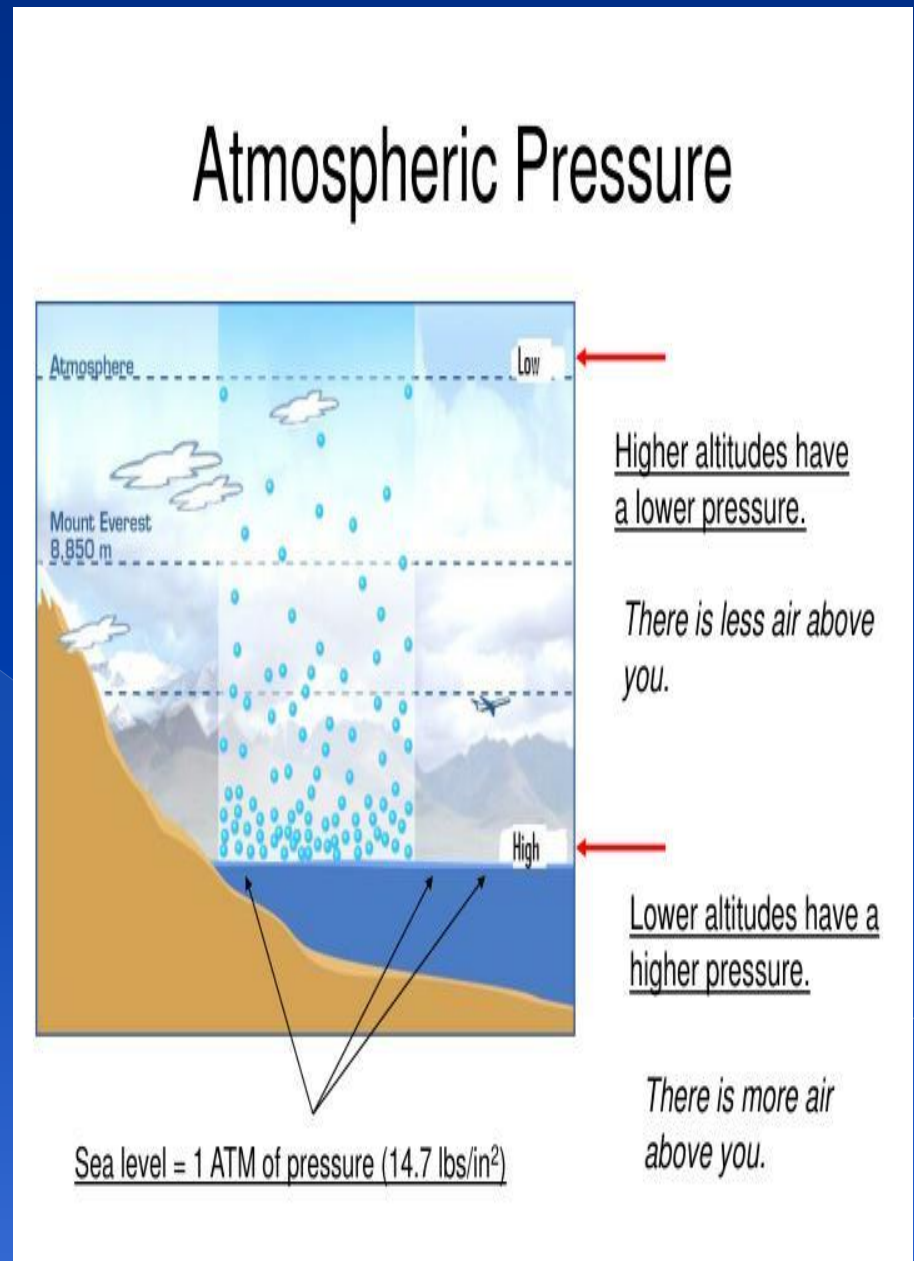


Atmospheric Pressure

- Pressure is defined as force per unit area.
- Atmospheric pressure is the force exerted at the earth's surface by a column of air above it.
- Pressure at a point acts uniformly in all directions.
- Units of pressure are:
 - ◆ millibar (mb).
 - ◆ hectopascal (hPa).
 - ◆ $1\text{mb} = 1\text{ hPa}$

- Clearly, the atmospheric pressure is highest at the surface of the Earth, i.e., at the sea level, and decreases as we go above the surface as the height of the column of air above decreases.

- The atmospheric pressure at sea level is called ***normal atmospheric pressure***.

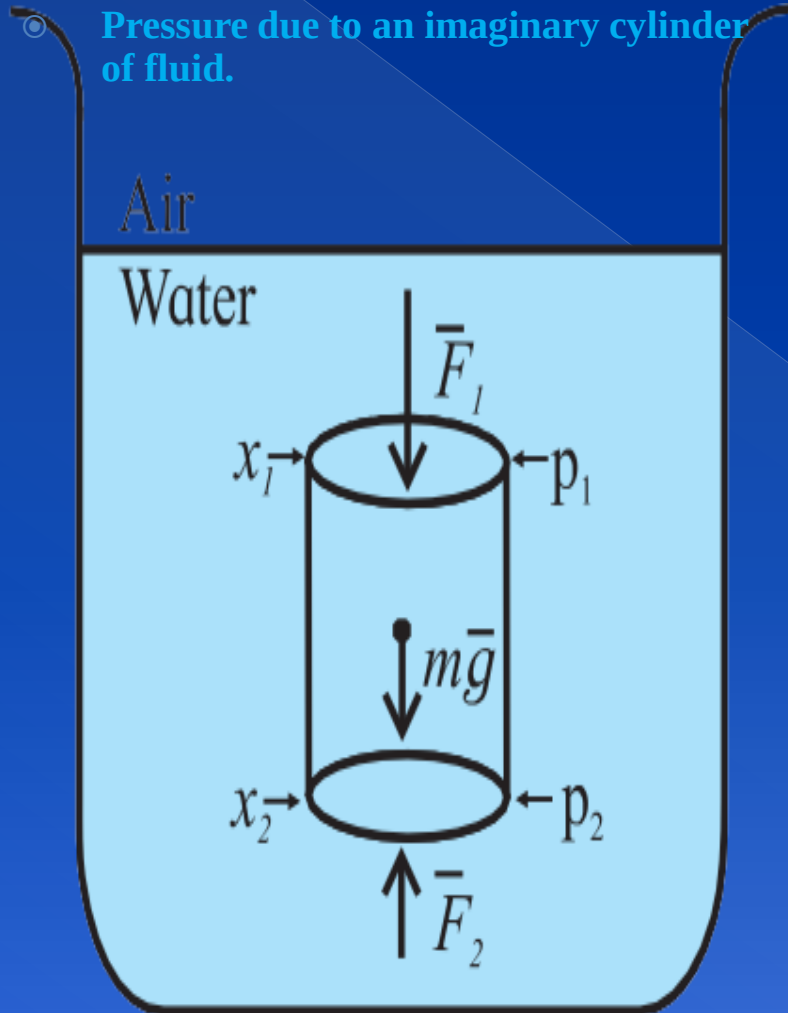


➤ The density of air in the atmosphere decreases with increase in height and becomes negligible beyond a height of about 8 km so that the height of air column producing atmospheric pressure at sea level can be taken to be 8 km.

➤ The region where gas pressure is less than the atmospheric pressure is called ***vacuum***.

➤ Perfect or absolute vacuum is when no matter, i.e., no atoms or molecules are present.

Absolute Pressure and Gauge Pressure:



- Consider a tank filled with water as shown in fig.
- Assume an imaginary cylinder of horizontal base area A and height $x_1 - x_2 = h$
- x_1 and x_2 being the heights measured from a reference point, height increasing upwards: $x_1 > x_2$. The vertical forces acting on the cylinder are

- Force F_1 acts downwards at the top surface of the cylinder, and is due to the weight of the water column above the cylinder.

- Force F_2 acts upwards at the bottom surface of the cylinder, and is due to the water below the cylinder.

- The gravitational force on the water enclosed in the cylinder is mg , where m is the mass of the water in the cylinder. As the water is in static equilibrium, the forces on the cylinder are balanced. The balance of these forces in magnitude is written as,

- $F_2 = F_1 + mg$ --- (2.3)

- p_1 and p_2 are the pressures at the top and bottom surfaces of the cylinder respectively due to the fluid. Using Eq. we can write

- $F_1 = p_1 A$, and $F_2 = p_2 A$ --- (2.4)

- Also, the mass m of the water in the cylinder can be written as,

- $m = \text{density} \times \text{volume} = \rho V$

- $\therefore m = \rho A(x_1 - x_2)$

- Substituting Eq. (2.4) and Eq. (2.5) in Eq. (2.3) we get,

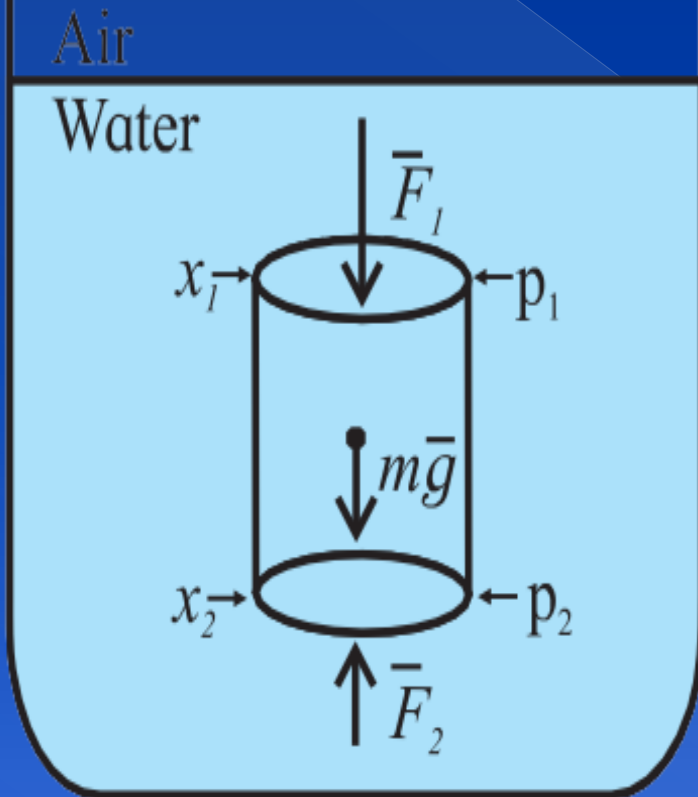
- $p_2 A = p_1 A + \rho A g(x_1 - x_2)$

- $p_2 = p_1 + \rho g(x_1 - x_2)$

- This equation can be used to find the pressure inside a liquid (as a function of depth below the liquid surface) and also the atmospheric pressure (as a function of altitude or height above the sea level).

To find the pressure p at a depth h below the liquid surface

- Pressure at a depth h below the surface of a liquid.



- let the top of an imaginary cylinder be at the surface of the liquid. Let this level be x_1 .
- Let x_2 be some point at depth h below the surface as shown in Fig. 2.8. Let p_0 be the atmospheric pressure at the surface, i.e., at x_1 .
- Then, substituting $x_1 = 0$, $p_1 = p_0$, $x_2 = -h$,

- To find the pressure p at a depth h below

- the liquid surface, let the top of an imaginary cylinder be at the surface of the liquid. Let this level be x_1 .

- Let x_2 be some point at depth h below the surface as shown in Fig.

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- Let p_0 be the atmospheric pressure at the surface, i.e., at x_1 . Then, substituting $x_1 = 0$, $p_1 = p_0$, $x_2 = -h$, and $p = p$ in Eq. (2.6) we get,

- $p = p_0 + h \rho g$...2.7

- The above equation gives the total pressure, or the absolute pressure p , at a depth h below the surface of the liquid.

- The total pressure p , at the depth h is the sum of:

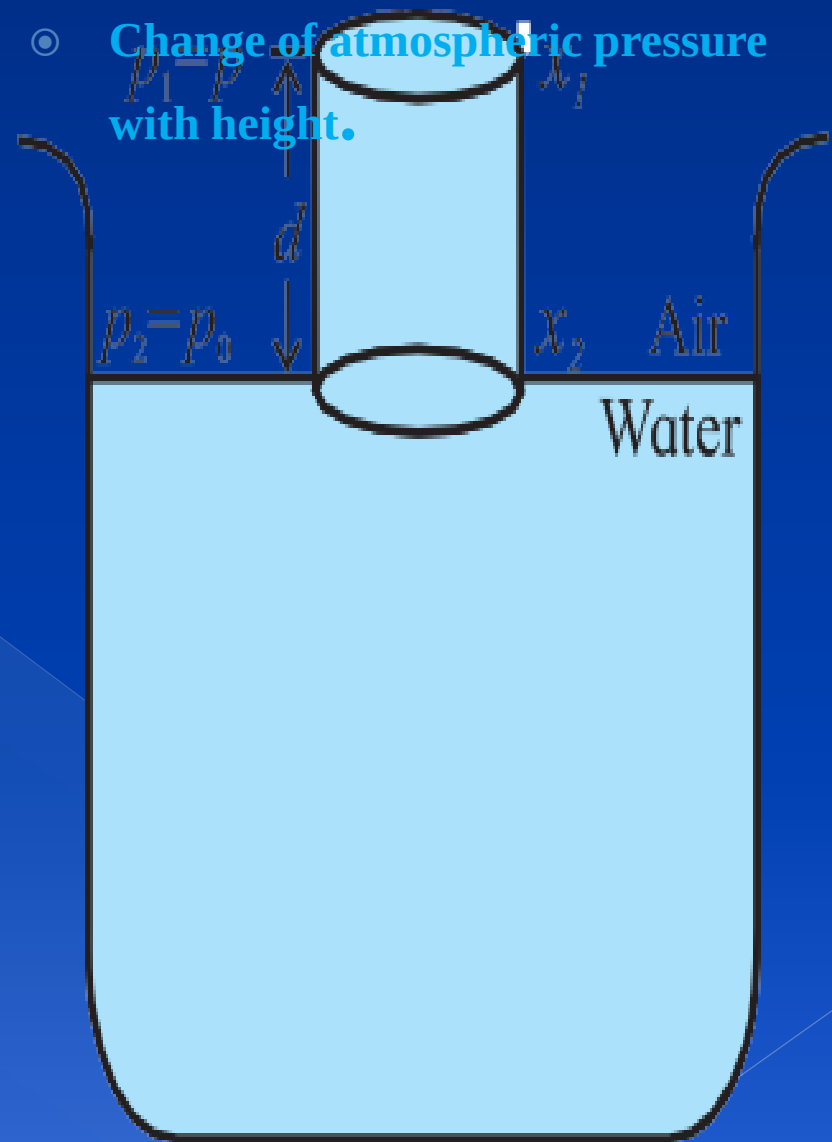
1. p_0 , the pressure due to the atmosphere, which acts on the surface of the liquid, and
2. $h\rho g$ the pressure due to the liquid at depth h .

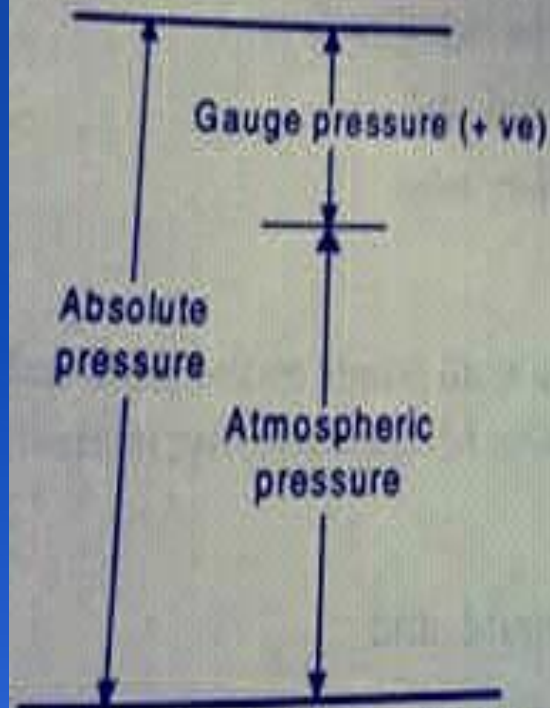
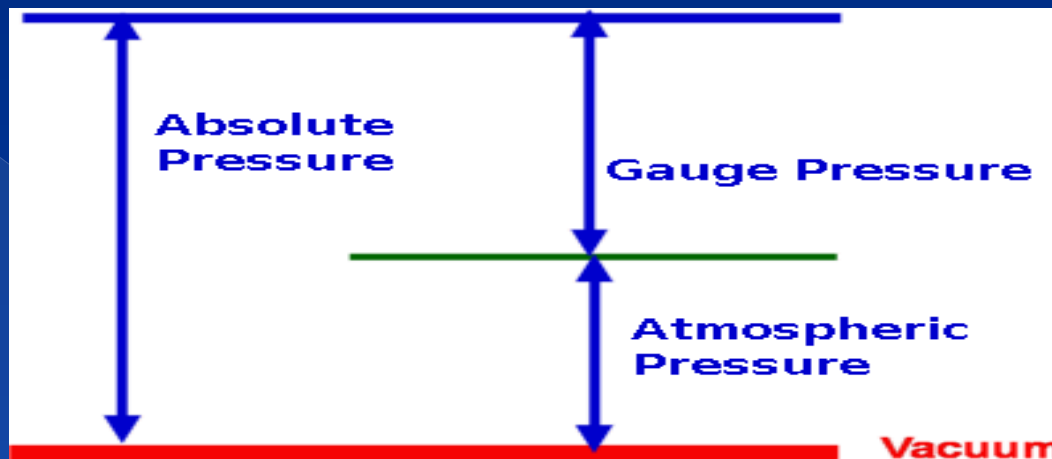
- ⊙ the difference between the absolute pressure and the atmospheric pressure is called the *gauge pressure*.
- ⊙ Using Eq. (2.7), gauge pressure at depth h below the liquid surface can be written as,
- ⊙ $p - p_a = h\rho g$ --- (2.8)

- To find the atmospheric pressure at a distance d above the liquid surface as shown in Fig. 2.9, we substitute $x_1 = d$, $p_1 = p$, $x_2 = 0$, $p = p_0$ and $\rho = \rho$ in Eq. (2.6) we get,

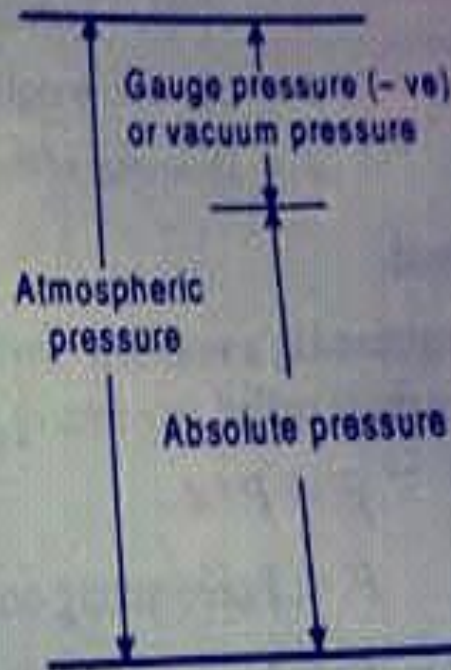
- $$p = p_0 - d\rho_{\text{air}}g \quad \text{---}$$

(2.9)





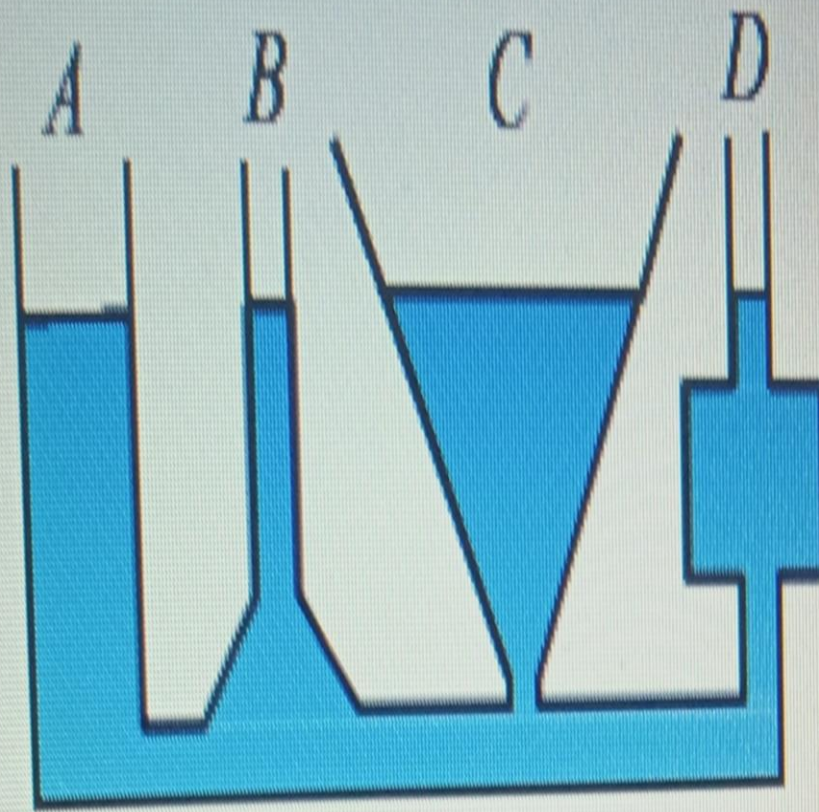
(a) Relation between absolute, atmospheric and gauge pressure.



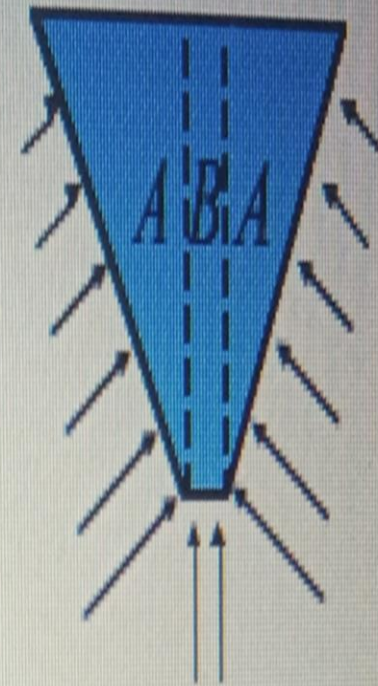
(b) Relation between absolute, atmospheric and vacuum pressure.

Hydrostatic Paradox:

- Consider the inter connected vessels as shown in Fig
- When a liquid is poured in any one of the vessels, it is noticed that the level of liquids in all the vessels is the same.
- This observation is somewhat puzzling.
- It was called 'hydrostatics paradox' before the principle of hydrostatics were completely understood.
- One can feel that the pressure of the base of the vessel C would be more than that at the
- base of the vessel B and the liquid from vessel C would rise into the vessel B. However, it is never observed. Equation 2.2 tells that the pressure at a point depends only on the height of the liquid column above it. It does not depend on the shape of the vessel. In this case, height of the liquid column is the same for all the vessels. Therefore, the pressure of liquid column in each vessel is the same and the system is in equilibrium. That means the liquid in vessel C does not rise in to vessel B.



(a)



(b)

Fig. 2.10: Hydrostatic paradox.

- ⦿ Consider Fig. 2.10 (b). The arrows indicate the forces exerted against the liquid by the walls of the vessel.
- ⦿ These forces are perpendicular to walls of the vessel at each point. These forces can be resolved into vertical and horizontal components.
- ⦿ The vertical components act in the upward direction.
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- ⦿ weight of the liquid in section B is not balanced and contributes the pressure at the base.
- ⦿ Thus, it is no longer a paradox!

- ⦿ Absolute pressure: Pressure which is related to reference pressure is known as absolute pressure.
- ⦿ Differential pressure: The difference between the two pressures is equal to the differential pressure.
- ⦿ Atmospheric pressure: Pressure related to the atmosphere that surrounds the earth is atmospheric pressure. This pressure is weather-dependent.
- ⦿ Gauge pressure: This is the difference between absolute pressure and atmospheric pressure.



PHYSICS

I'm Less Concerned With How He Got Up There
Than I Am About What's Holding Those Bricks Up